

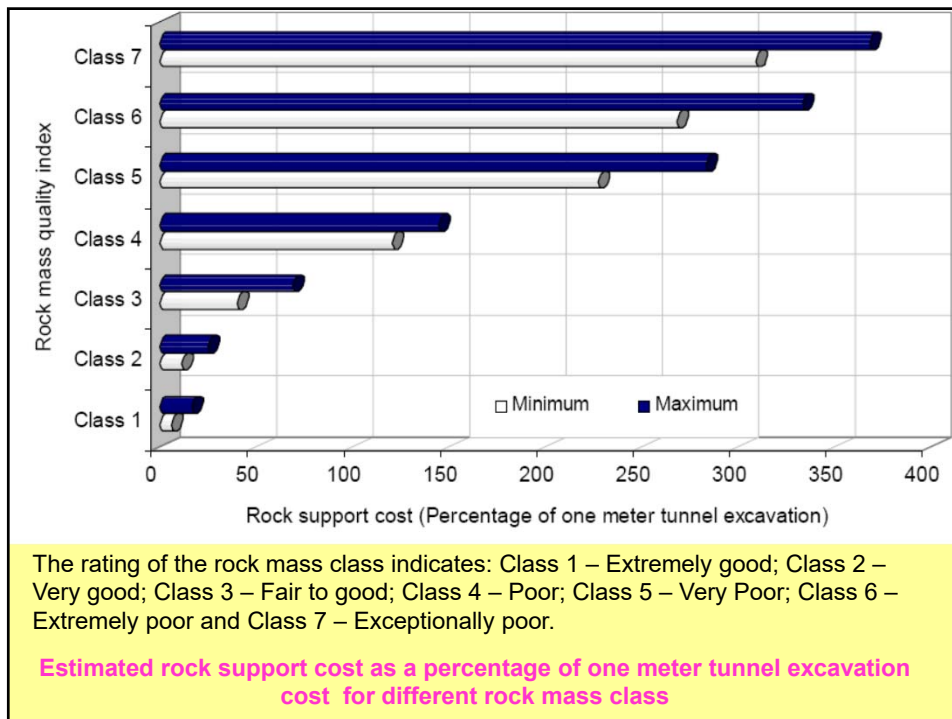
GEOLOGICAL AND GEOTECHNICAL SITE CHARACTERIZATION FOR ROCK ENGINEERING PROJECTS

Geological and Geotechnical site characterization

The knowledge of *geological and geotechnical site characterization* of rock mass is necessary for *proper design of excavation* can result in potentially large *cost saving, stable* and *safe* working environment.

Geological and Geotechnical site characterization

The characterization of rock mass quality is very essential during pre-construction phase geological investigations since it is the only way to estimate required tunnel rock support and construction time. Without estimating the quantity of rock support and construction time it is not possible to evaluate the economic viability of a tunnel project during planning.



Site Characterization

An important question which requires careful consideration at the start of any tunnel and underground excavation for rock engineering project is: **What constitutes a realistic geotechnical site characterization for that project and to what extent can such a site investigation be expected to minimize the unknowns which could give rise to tunnelling problems and stability of the excavation?**

Site Characterization

The prime purpose of any geotechnical site characterization should be to obtain the maximum amount of information on

- **rock characteristics,**
- **structural systems and**
- **groundwater conditions.**

Site Characterization

This information is important to the underground excavation designer in that it should enable him to anticipate the behaviour of the rock surrounding the opening and the type of support as well as the size, shape and orientation of all the openings that can be safely and economically excavated in a particular rock mass.

Site Characterization

The information is also important to the persons carrying out excavation in that

- ✓ it should provide him/her with a basis for establishing the optimum excavation method
- ✓ type of services which he/she will require in order to meet the construction schedules.

Site Characterization

Geotechnical site characterization engineering (commonly addressed as Geotechnical engineering) deals with the whole spectrum of natural geological materials. Geological and weathering processes have resulted in a wide range of natural materials ranging from low strength soils to high strength rocks.

Site Characterization

Geotechnical engineering is one of the tools that the excavation is encouraged to apply in the continuing endeavour to achieve safe, cost effective underground openings.

GEOTECHNICAL SITE CHARACTERIZATION

The requirement to rehabilitate extensive areas of a openings in rock, due primarily to a failure to address the prevailing geotechnical issues, exposes the workforce to potentially hazardous ground conditions.

GEOTECHNICAL SITE CHARACTERIZATION

A number of fundamental geotechnical characteristics of the ground/rock mass effectively determine ground movement and behaviour in and around an underground excavation:

- **Geological structure/discontinuities**
- **Rock stress**
- **Engineering properties of the ground/rock mass**
- **Groundwater**

GEOTECHNICAL SITE CHARACTERIZATION

Consequently, the excavation operator should develop a **geotechnical model** that quantitatively defines each of the above ground characteristics in all areas of the excavation site. The nature and form of the geotechnical model can be considered as a **database that allows** relevant and truly representative information to be readily distilled **for critical design purposes such as support and planning requirement.**

GEOLOGICAL DATA

It should describe the **regional and site geology** and provide clear and unambiguous information on location and extent of **different material types**, i.e. lithology, the degree and type of alteration or weathering, which can significantly change material properties.

GEOLOGICAL DATA

•**REGIONAL GEOLOGY**: It requires an understanding of geological events that led to the formation of

- Regional geological structure
- Lithology
- Topography
- Morphology
- Regional stress field

•**SITE GEOLOGY** : It should represent a broader view of the geology of the excavation site, focussing on the engineering aspects.

- Type of large mass (soil/rock)
- Lithology
- Degree of Alteration and weathering of rockmass
- Geological structures and its characteristics

GEOLOGIC STRUCTURE

Geological structural refers to naturally occurring breaks in the rock such as **bedding planes**, **joints** and **faults**, which are generally termed **discontinuities**.

A discontinuity is a plane or surface that marks a change in physical or chemical characteristics in rock material. Discontinuities can be **classified** into **mechanicals** or **integrals**.

GEOLOGIC STRUCTURE

A **mechanical discontinuity** denote a plane of **physical weakness**, this means that the tensile strength perpendicular to this surface or the shear strength along it are **lower than those of the surrounding material**.

Contrarily, an **integral discontinuity** is as **strong as the surrounding material**. Integral discontinuities can become mechanical due to weathering or chemical reactions that develop a change in mechanical properties.

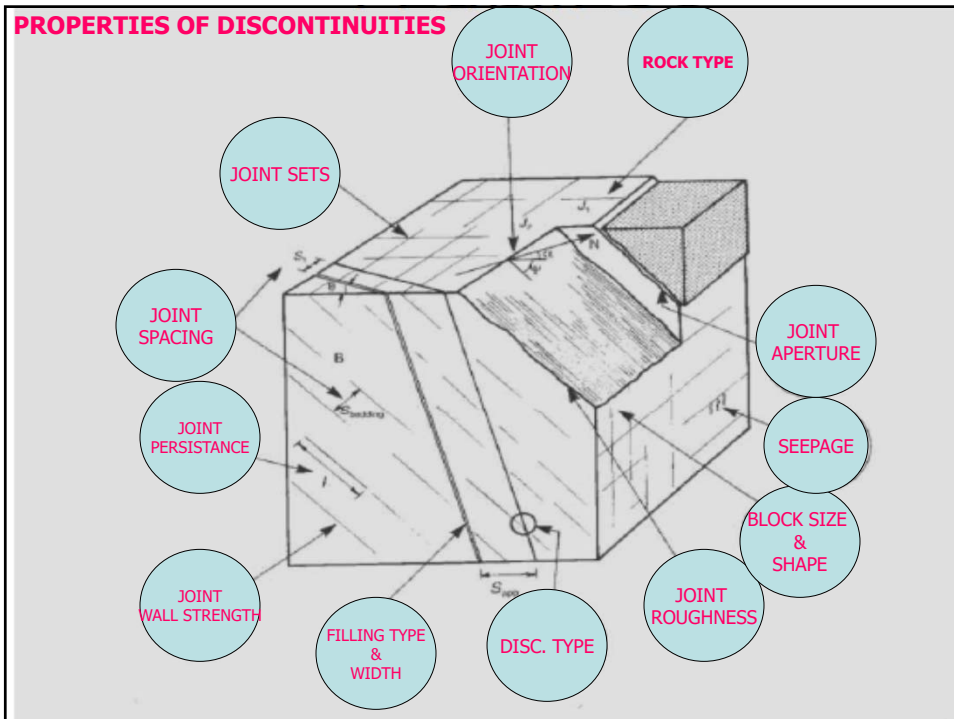
GEOLOGICAL DISCONTINUITIES

In geotechnical engineering the term geological discontinuities refers to all the natural planes of weakness in the rock mass that pre-date any excavation activity and includes: joints, faults, shears, bedding planes, foliation and schistosity. Discontinuities play a predominant role in influencing the stability of rock masses, both for rock slopes as well as in underground excavations.

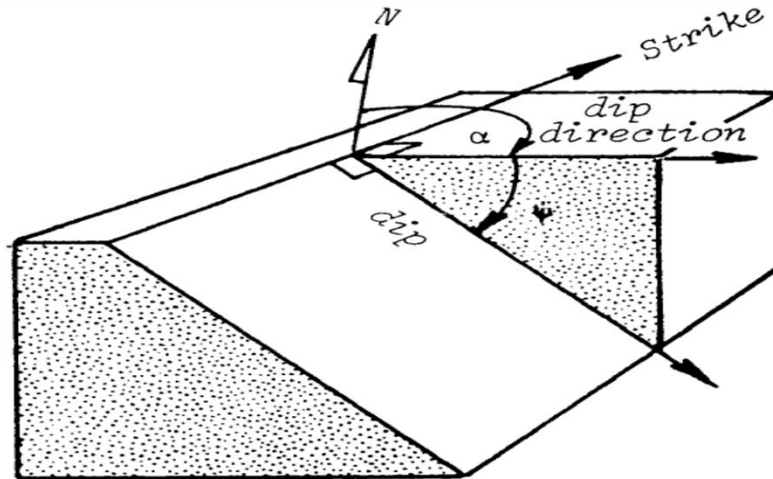
GEOLOGICAL DISCONTINUITIES

Geological discontinuities can have a range of characteristics including:

- ♦ Orientation - usually specified by dip angle and dip direction
- ♦ Spacing
- ♦ Persistence or continuity
- ♦ Roughness
- ♦ Wall strength
- ♦ Aperture
- ♦ Filling
- ♦ Seepage
- ♦ Number of sets



Orientation: It refers the attitude of the discontinuities in space. It is described by dip direction and dip of the line of steepest inclination in the plane of the discontinuity.



Orientation

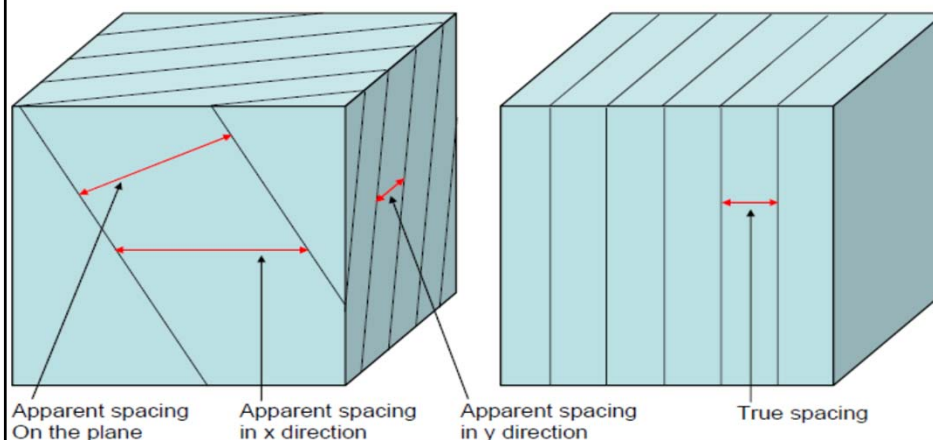
Orientation of a discontinuity is described by its dip and dip direction or its dip and strike. The orientation of major joint set relative to an opening in rockmass largely controls the possibility of unstable conditions or excessive deformations developing. The mutual orientation of discontinuities will determine the shape of the individual blocks and beds comprising the rock mass.

Orientation

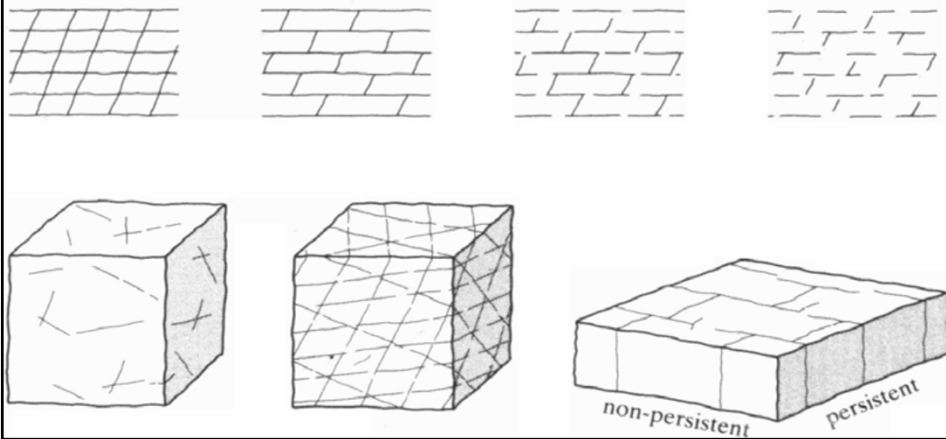
Dip or dip angle represents the degree of inclination. It is the acute angle between the plane and the horizontal plane. It is also the acute angle between a line with maximum dip in the inclined plane and its horizontal projection. Dip angle is generally expressed by an acute angle between 0° and 90° .

Dip direction represents the facing direction. It is the bearing measured clockwise from the north (0°) of the line with maximum dip in the inclined plane. Dip direction is generally expressed by a direction angle of 0° to 360° .

Spacing: Spacing can be defined as the perpendicular distance between adjacent discontinuities. Normally refers to the mean or modal spacing of a set of joints.



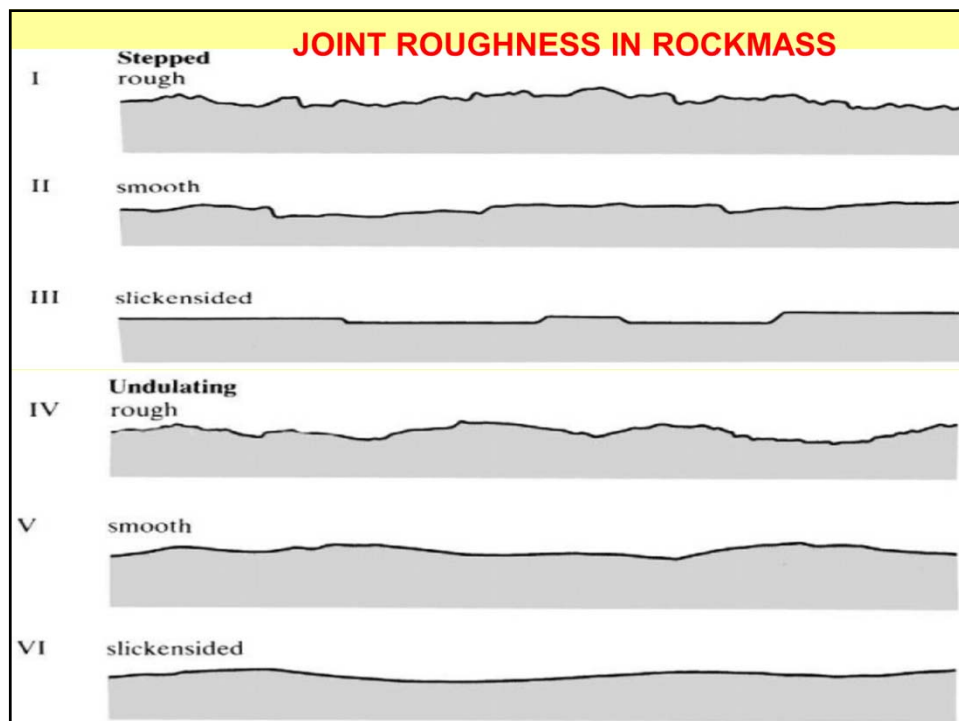
Persistence: Discontinuity trace length as observed in an exposure may give a crude measure of the areal extent or penetration length of a discontinuity. Termination in solid rock or against other discontinuities reduces the persistence.

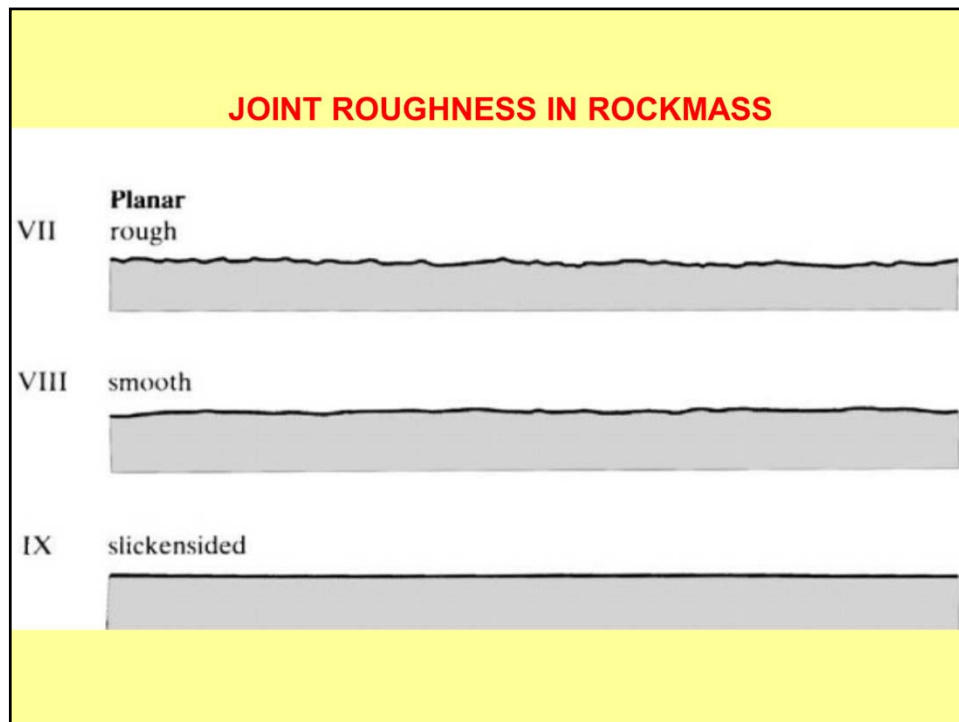


Roughness: The roughness may be defined as the inherent surface roughness and waviness relative to the mean plane of a discontinuity. Depending on nature of joint roughness in rock mass, it has been classified into nine types. Barton introduced the Joint Roughness Coefficient (JRC) as another method to quantify joint roughness in rock mass.

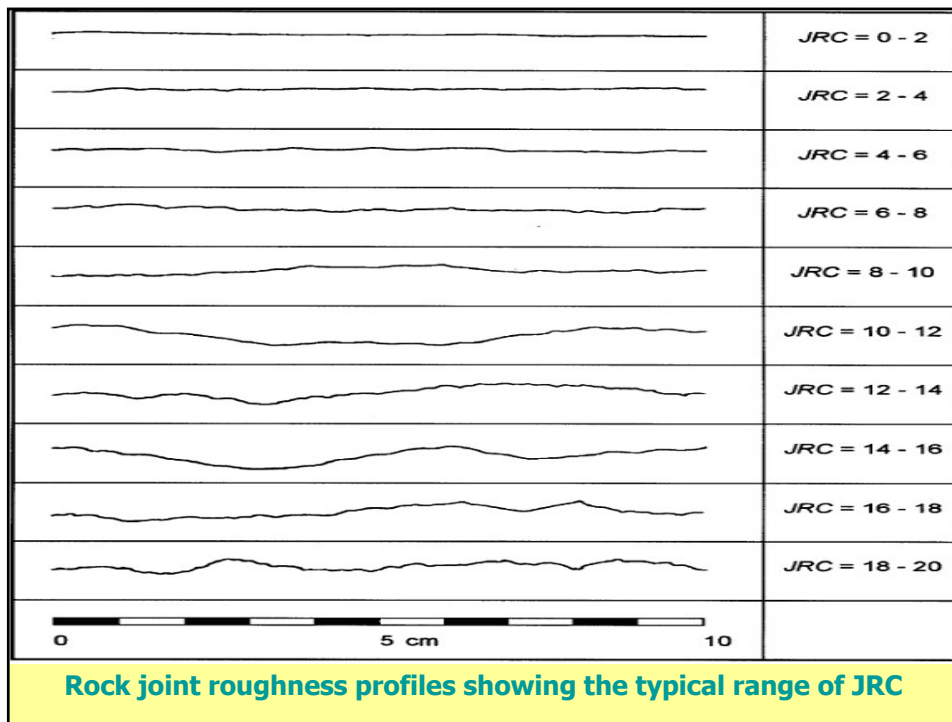
The surface roughness of geological discontinuities plays an important role in influencing the stability of rockmass. As Joint roughness has a strong influence on the engineering behaviour of rock mass.

Joint roughness has a direct impact on both hydraulic conductivity of fractures and the shear strength of rock joints





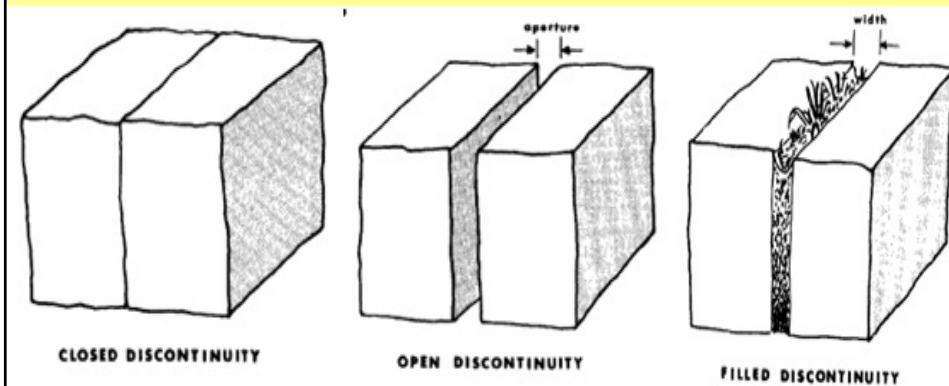
JOINT ROUGHNESS IN ROCKMASS		
	DESCRIPTION	JRC
	rough	20
	smooth	14
	slickensided	11
	Stepped	
	rough	14
	smooth	11
	slickensided	7
	Undulating	
	rough	2.5
	smooth	1.5
	slickensided	0.5
	Planar	



Wall Strength: This is referred to the equivalent compression strength of the adjacent rock walls of a discontinuity. It may be lower than rock block strength due to weathering or alteration of the walls. It is an important component of shear strength, if rock walls are contact.

Joint wall strength can be estimated by a Schmidt hammer. With the Schmidt hammer number, uniaxial compressive strength can be estimated.

Aperture: The aperture is the perpendicular distance between adjacent rock walls of a discontinuity in which the intervening space is air or water filled.

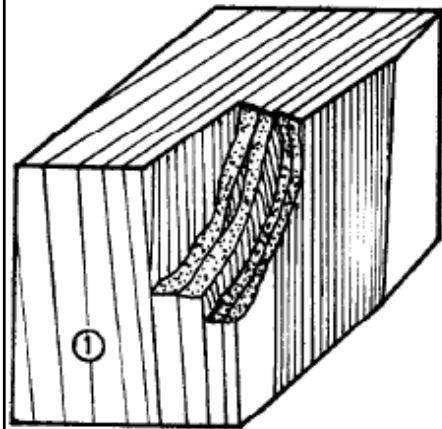


Fillings: It is the material that separates the adjacent rock walls, of discontinuity and that is usually weaker than the parent rock. Typical filling materials are sand, slit, clays, breccia, gouge, and mylonite. Also includes thin mineral coating and healed discontinuities quartz and calcite veins.

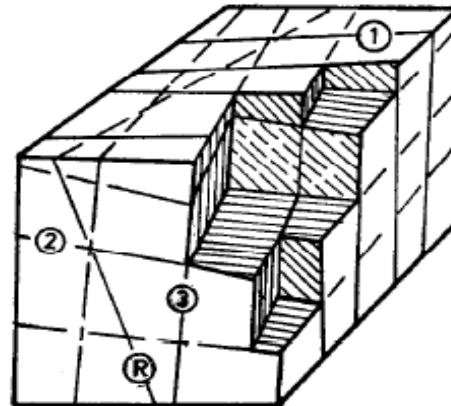
Seepage: Seepage is referred to the water flow and free moisture visible in individual discontinuities or in the rock mass as a whole.

Block Size: Rock block dimensions, resulting from the mutual orientation of intersecting joint sets and resulting from the spacing of the individual sets. Individual discontinuities may further influence the block size and shape.

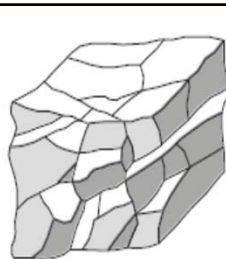
Number of Sets: The discontinuities occur as sets. In each set, the discontinuities are parallel and in such along a number of sets may occur.



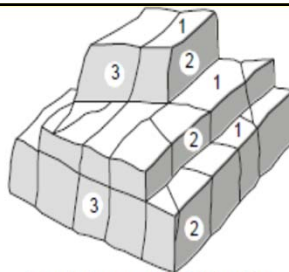
ONE JOINT SET



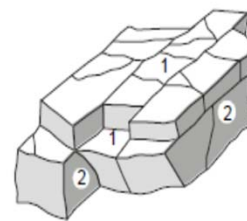
THREE JOINT SETS



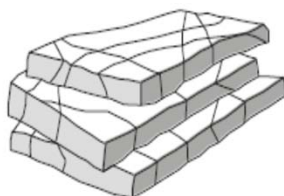
Polyhedral blocks



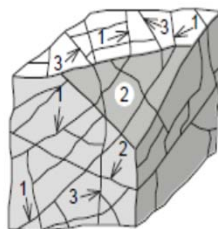
Equidimensional blocks



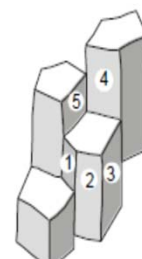
Prismatic blocks



Tabular blocks

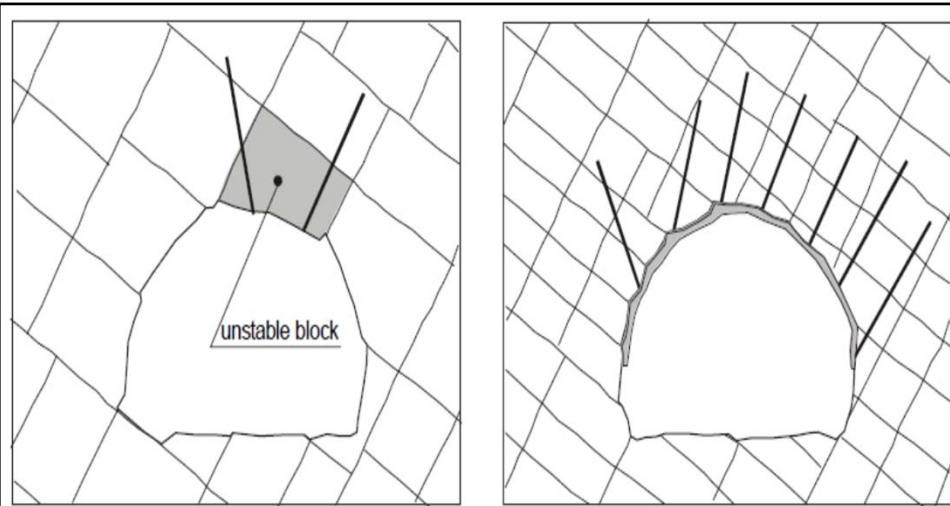


Rhombohedral blocks



Columnar blocks

Examples of block shapes or the jointing pattern



Block size has a main influence on stability in underground openings often being supported.

Weathering

Reduction of rock strength due to weathering reduces the shear strength of discontinuities. Weathering will also reduce the shear strength of the rock mass due to the diminished strength of the intact rock. Weathering categories range from fresh rock to residual soil. Weathering of rock takes the form of both disintegration and decomposition. Disintegration is the result of environmental conditions such as wetting and drying, freezing and thawing that break down the exposed surface layer.

The mechanical properties of the rock mass is obviously influenced by the presence of joint sets and the number of joint sets. More joint sets provide more possibilities of potential slide planes for rock wedges or blocks to slide and fall.

DATA COLLECTION TECHNIQUES

Several tools and techniques are available to the Geotechnical operator for data collection for geotechnical design and stability analysis of excavation in rock. These include

- Geophysical methods for subsurface geological data
- Exposed rock surface mapping for surface geological data
- Geotechnical drilling and core logging for subsurface geological data
- Field and laboratory testing for engineering properties of intact rock material and rock mass
- Hydrological test
- Insitu stress measurement

GEOPHYSICAL METHODS

The high cost of sub-surface exploration by core drilling or any existing surface and underground excavations , results in that the use of these investigations is generally limited. Geophysical methods can often be used to supplement the information from such explorations.

GEOPHYSICAL METHODS

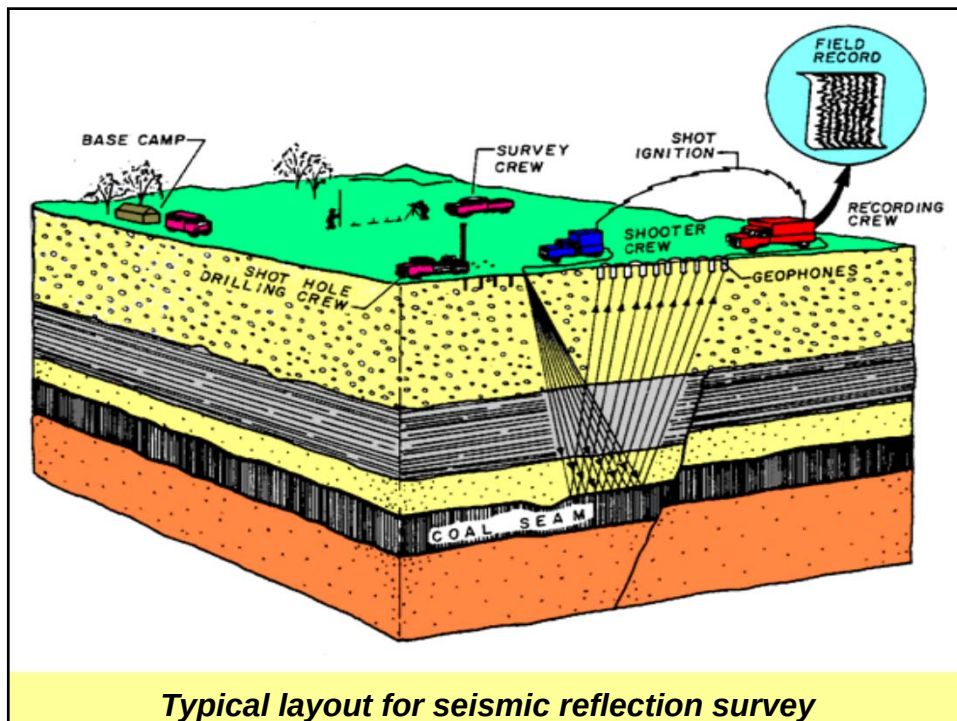
Geophysical methods are often used in the reconnaissance or preliminary stages of a site investigation program to provide such information as demarcation of major lithological units and major structural features such as fault, fracture zones etc. Geophysical surveys provide a continuous profile of subsurface conditions and this information can be used as a fill-in between drill holes.

The results obtained from geophysical measurements are usually not sufficiently accurate.

GEOPHYSICAL METHODS

At early stages of a project, the following geophysical methods can be used for obtaining geological information:

- Seismic reflection measurements
- Seismic refraction measurements
- Cross hole tomography measurements
- Resistivity measurements
- Ground Penetrating Radar (GPR)
(Electromagnetic measurements)
- Gravity measurements



Ground Penetrating Radar (GPR)

GPR based methods have allowed to detect and map internal discontinuities and derive a interpretation of a rock mass internal discontinuities network.

GPR uses high energy electromagnetic radiation (EMR) in a microwave range with frequencies between 10MHz-2.5GHz. The interaction of those waves with rock depends on its dielectric constants. If the radar waves meets the boundary of two materials with a different permittivity a part of the energy reflected and other part is transmitted.

Ground Penetrating Radar (GPR)

The velocity of radar wave in a material varies with its dielectric permittivity and attenuation increase with increasing frequencies (therefore with increasing wavelength as well).

GPR equipment consists principally of a control unit, a transmitter and a receiver. A computer is used for data collection.

Ground Penetrating Radar (GPR)

GPR has demonstrated to be a suitable geophysical method to detect internal discontinuities in a rock mass. Depending on the survey requirements, it can reach the required vertical and horizontal resolution and depth penetration.

Multi-reflections and 3D geometric effects can be also possible in discontinuities with a roughness of strong amplitude.

EXPOSED ROCK SURFACE MAPPING

Surface outcrops must be utilised to obtain information on the engineering properties and structure of the rock mass. Measurements may be made on natural outcrops or on faces exposed by surface excavations. In some projects, any existing pit slopes and underground excavations provides an invaluable source of data.

At a regular interval surface mapping of current the excavation must be carried out once excavation has commenced for up dating previous data.

Data collection from exposed rock surfaces, particularly the data on type of geological structure, aperture, persistence, filling, orientation, spacing, length and surface waviness (roughness) of geological structures, can also be carried out.

EXPOSED ROCK SURFACE MAPPING

1-Traditional methods surface mapping method

The approaches used for mapping exposures may be of three main types:

- **Spot mapping**
- **Lineal mapping**
- **Areal mapping**

2-Remote sensing technique

- **Terrestrial Laser Scanning Method**
- **Satellite Remote Sensing**

EXPOSED ROCK SURFACE MAPPING

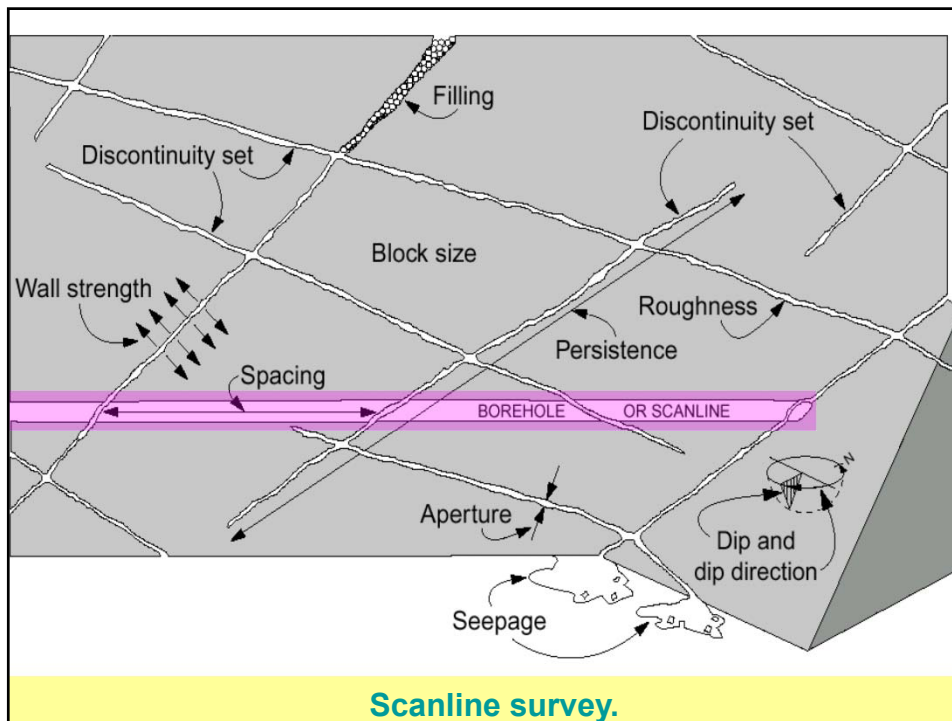
Spot mapping in which the observer selectively samples only those discontinuities that are considered to be important [example- *Slope Stability Probability Classification (SSPC) system*].

Lineal mapping in which all discontinuities intersecting a given sampling line are mapped [example- *scanline survey*].

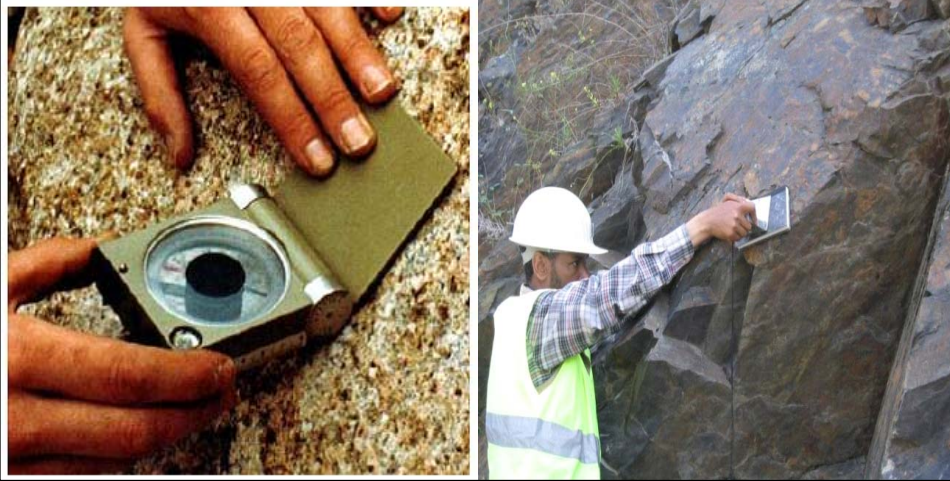
Areal mapping in which all discontinuities within a selected area of the face, often called a window, are mapped. This method can reduce some of the biases in mapping.

EXPOSED ROCK SURFACE MAPPING

The basic technique used in mapping surface or underground exposures is the **scanline survey**. **Scanline** is a one dimensional discontinuity sampling technique. A line is located on the rock face and discontinuity planes that intersect the scan line are registered with their properties (i.e. location, orientation, trace or semi trace length, persistency, roughness, and infill). A scan-line survey provides statistical information for engineering design purposes; however scan-line is not a standardized method. Additionally, the method can introduce sampling bias in each measured parameter and corrections have to be used to compensate it.

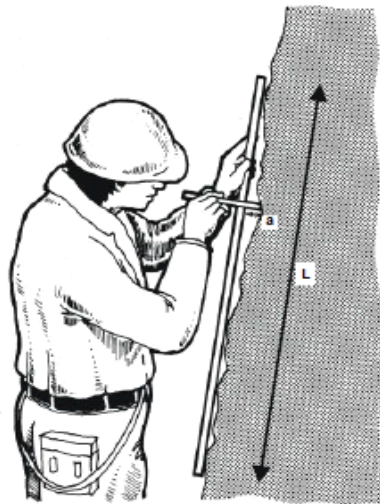


The most convenient way to measure joint orientation is from accessible outcrops or exposed faces of slope cuts or underground excavation. The measurements can be made by a geological compass, clinometer or Brunton compass that yield readings of dip direction (bearing) and dip angle (inclination), as shown in Figure [REDACTED]. More recent gadgets such as Android phones with applications like Rocklogger can be very handy and economic tools for any engineer working in the field.



Measurement of Roughness

$$u = \frac{\text{measured max. amplitude (}}{\text{measured length along joint}}$$



EXPOSED ROCK SURFACE MAPPING

Data collection from exposed rock surfaces, can also be carried out using 3D digital photogrammetric techniques. With these techniques the data can be gathered remotely and accurately from areas where access is difficult or unsafe. Such techniques permit accurate low-cost mapping of surface at a rapid rate. The data collected by these techniques can be downloaded into software and used in real time for mine planning, design, operating stability analysis purposes.

Measurement of discontinuity geometrical features

Feature	Measurement Method	Outcrop Mapping	Core Logging	Borehole Logging
Discontinuities type	Visual	good	good	medium
Orientation	Compass-clinometer	good	medium	good
Spacing	Measuring tape	good	good	medium
Persistence	Measuring tape	good	poor	poor
Roughness	Profile gauge	good	medium	poor
Wall strength	Schmidt hammer	good	medium	poor
Aperture	Scale or feeler gauge	good	poor	poor
Filling	Visual	good	poor	poor
Seepage	Timed observation	good	poor	good
Number of joint sets	Hemispherical projection	good	medium	poor
Block size	3-D fracture frequency	good	poor	poor

Terrestrial Laser Scanning

Terrestrial Laser scanning is a new method for surveying tasks, which allows to acquire easy and fast complex geometric characterization of discontinuities based on the interpretation of the visible surface of a rock mass outcrop include image analysis, digital photogrammetry and total station.

The rock mass exposure was scanned in a single survey using a time based 3-D laser scanner.

Terrestrial Laser Scanner is a remote sensing technique which has shown a great potential to obtain a large quantity of accurate geometric data.

Terrestrial Laser Scanning

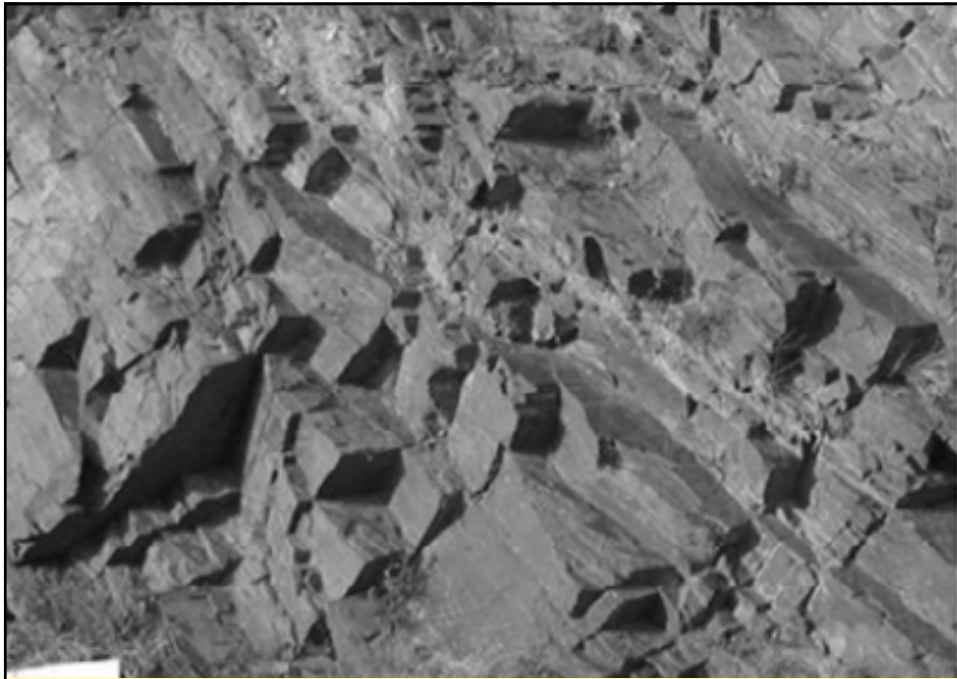
During a TLS survey, a laser beam emits repeated pulses that are reflected by the rock face. These measurements can be translated into xyz-coordinates, using either the two way time of flight for each received pulse or the amplitude modulated continuous wave principle. This results in the acquisition of a dense 3D point cloud of the rock surface with a high spatial resolution (5 – 10 mm), which represents the surface shape at a very high detail. The xyz data can be complemented with the intensity value of each returned pulse and color information extracted from digital imagery.

Terrestrial Laser Scanning

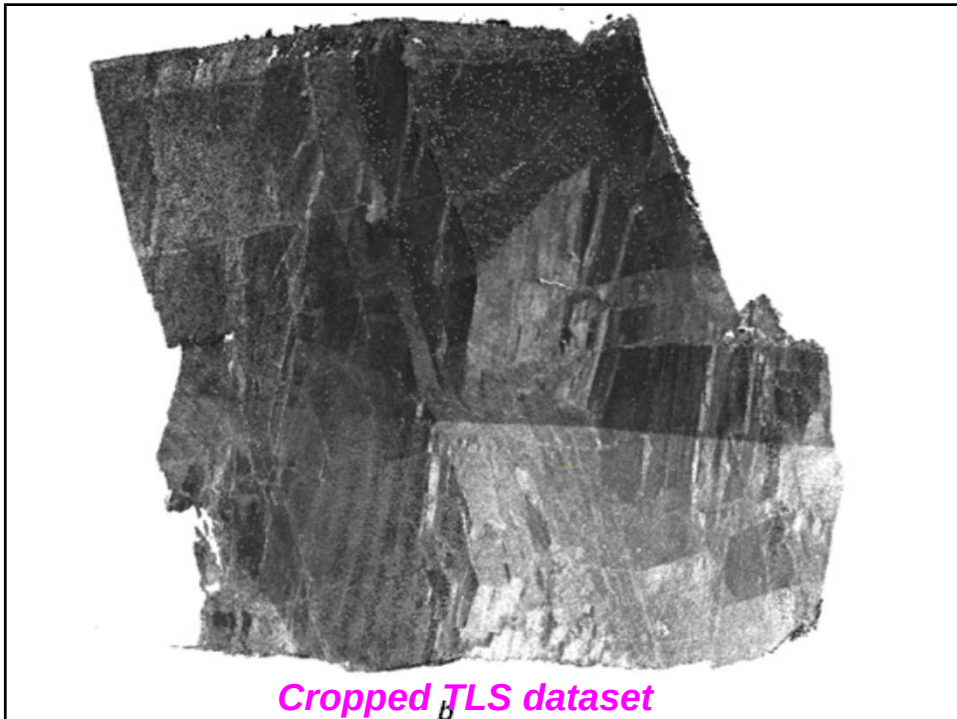
Advantages of laser scanning like:

- Non contact measurement
- High accuracy: mm to cm, depending on range, laser features, reflectivity
- Long range-max measurement range: up to 2-3 km
- Fast data acquisition: emission-detection of laser pulses ($10^4 - 10^5$ measurements/sec) and point clouds in xyz + optical RGB digital imaging

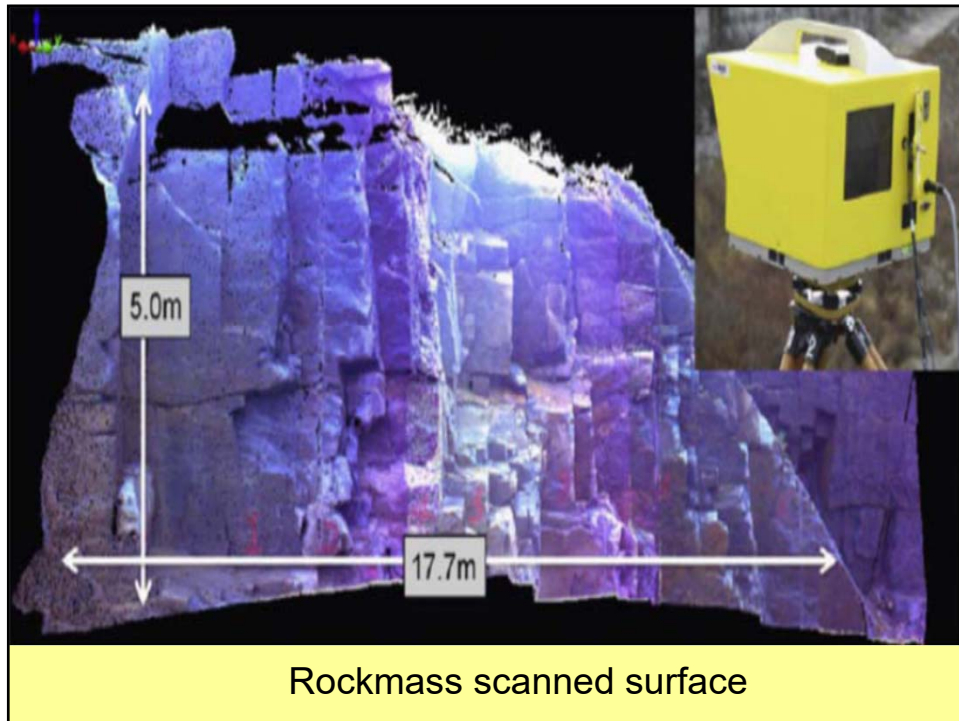




Rockmass scanned surface



Cropped TLS dataset



Geotechnical drilling and core logging

Prior to the commencement of excavation, much needed subsurface information for excavation stability analysis, within acceptable levels of confidence, can be obtained by core drilling and logging. By advance planning and scoping, the core drilling programs primarily aimed at mineral exploration and resources evaluation purposes can be used to extract geotechnical information.

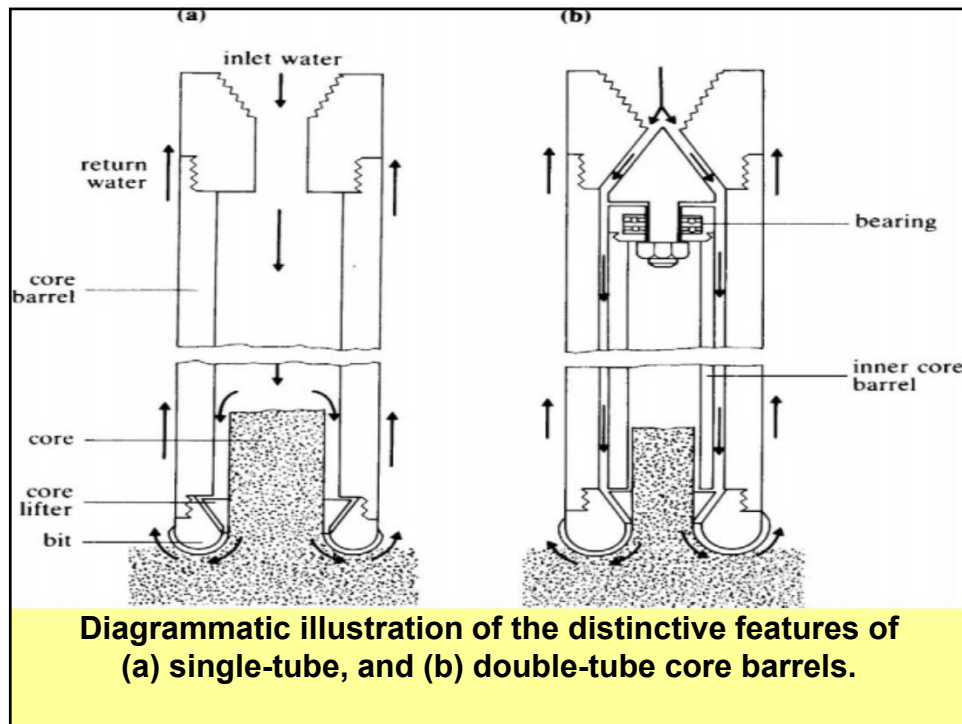
Geotechnical drilling and core logging

The purpose of a core drilling investigation can be to:

- Confirm the geological interpretation.
- Obtain information on the rock types and their boundaries in the rock mass.
- Obtain more information of the rock mass structure.
- Study ground water conditions.
- Provide material (samples) for rock mechanics testing and petrographic analyses.

Geotechnical drilling and core logging

Diamond drilling machine: Rotary drill machine with suitable diamond drilling bit and a hydraulic feed drilling machine is essential to ensure high core recovery. Drill machine should have double tube core barrel instead of single tube core barrel for obtaining complete core. When a single-tube core barrel is used, the core may be damaged by the rotating barrel and by the circulating water to which it is fully exposed. In a double tube core barrel, the core is retained in an inner barrel mounted on a bearing assembly.



Geotechnical drilling and core logging

Core drilling programs primarily executed for geotechnical data collection will also be required to gain an adequate understanding of the subsurface conditions before the commencement of excavation.

Obviously, the number of geotechnical holes required for a particular project will depend on the level and reliability of already available geological and geotechnical information of the site, the complexity of site geology and the size and operating life of the project.

Geotechnical drilling and core logging

Components of the drilling could include core orientation to obtain structural geology information, and installation of piezometers to measure ground water levels, and possibly measure permeability. Rock strength testing could comprise laboratory testing of drill core to determine the friction angle of discontinuities, rock strength tests and slake durability tests.

Geotechnical drilling and core logging

Core logging: The final stage in the geotechnical drilling process is the recording of the information obtained from the core. Here again, the value of the entire expensive exercise can be put at risk by the use of poor techniques or insufficient care. Generally, the structural or geotechnical logging of the core is carried out by specially trained operators in a location removed from the drilling site. The log so obtained is additional to the normal driller's or geologist's log.

CORE DRILLING AND LOGGING

Core samples recovered from boreholes can be logged by traditional methods of direct observation or using digital photography. Suitable software is now available for the analysis of data recorded.

Boreholes can also be logged using downhole cameras and downhole geophysical techniques.

Geotechnical drilling and core logging

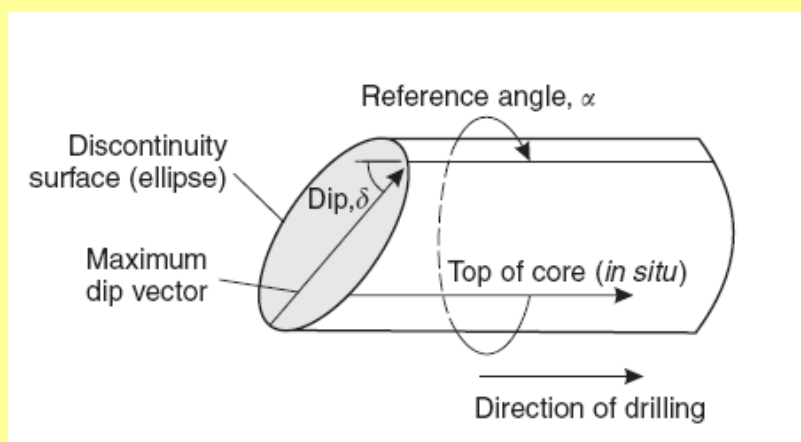
Traditional methods core logging:

Recording the properties of the recovered drill core involves rock description, the properties of the discontinuities and their orientation with respect to the core axis. Measurements can also be made of the Rock Quality Designation (RQD), fracture index and core recovery, which are indicative of the rock mass quality, as described.

Geotechnical drilling and core logging

Core orientation: If the fullest possible structural data are to be obtained from the core, it is essential that the core be oriented correctly in space. Not only must the trend and plunge of the borehole axis be measured (the trend and plunge of a line are analogous to the dip direction and dip of a plane), but the orientation of the core around the full 360° of the borehole periphery must be recorded. If this is not done, then the true orientations of discontinuities intersected by the borehole cannot be determined.

Measurement of orientation of discontinuity in oriented drill core.



Geotechnical drilling and core logging

The use of drill rig instrumentation, sometimes known as **measurement while drilling (MWD)**, can greatly improve the quantity and quality of the geotechnical data obtained from non-coring rotary drilling as well as from coring drilling.

Geotechnical drilling and core logging

Downhole logging: The logging of core may be supplemented by logging of the borehole walls with a range of downhole tools such as borehole cameras, television cameras and geophysical tools. Optical imaging may be enhanced by the use of digital imaging software to process the signals received from a downhole scanner.

Plotting and analysis of discontinuity orientation data

Field data may be plotted using stereonet.

A stereonet is the projection of planes and a 3-dimensional reference sphere through which they might pass, to a 2-dimensional representation. This allows the orientations of planes in space to be accurately represented and easily visualized as illustrated by Figure .

Plotting and analysis of discontinuity orientation data

There are several types of projections.

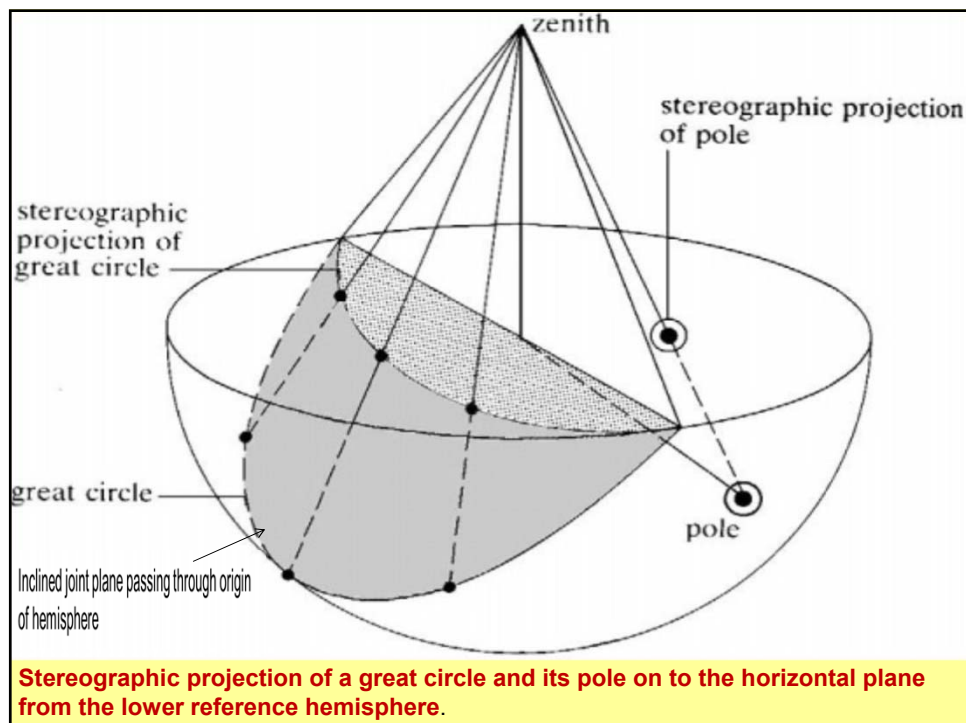
The two projections most commonly used by geologists for structural analyses are

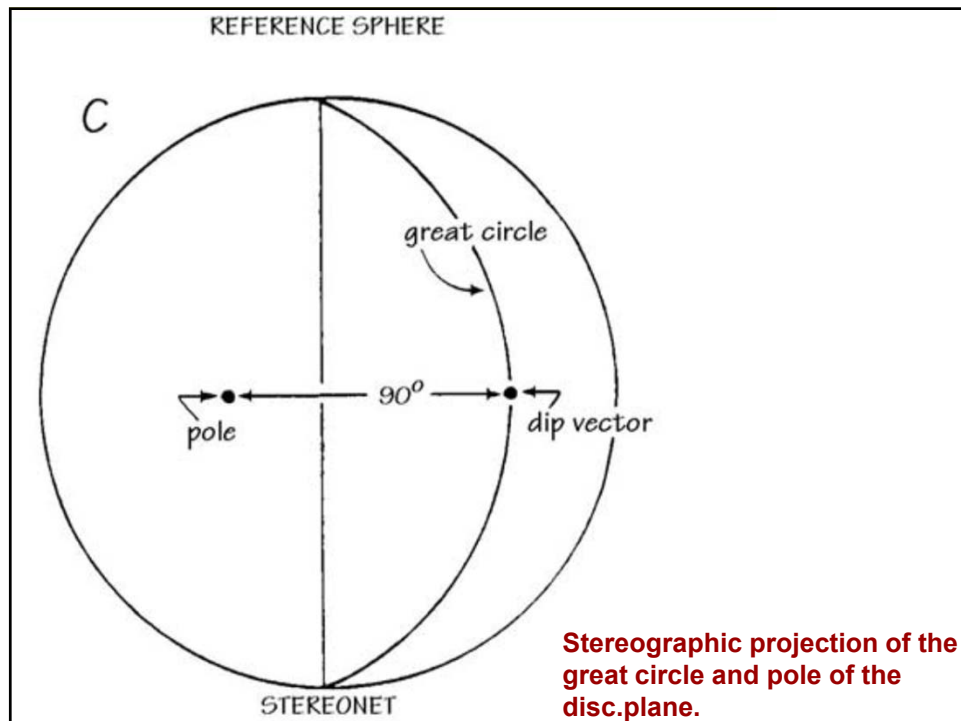
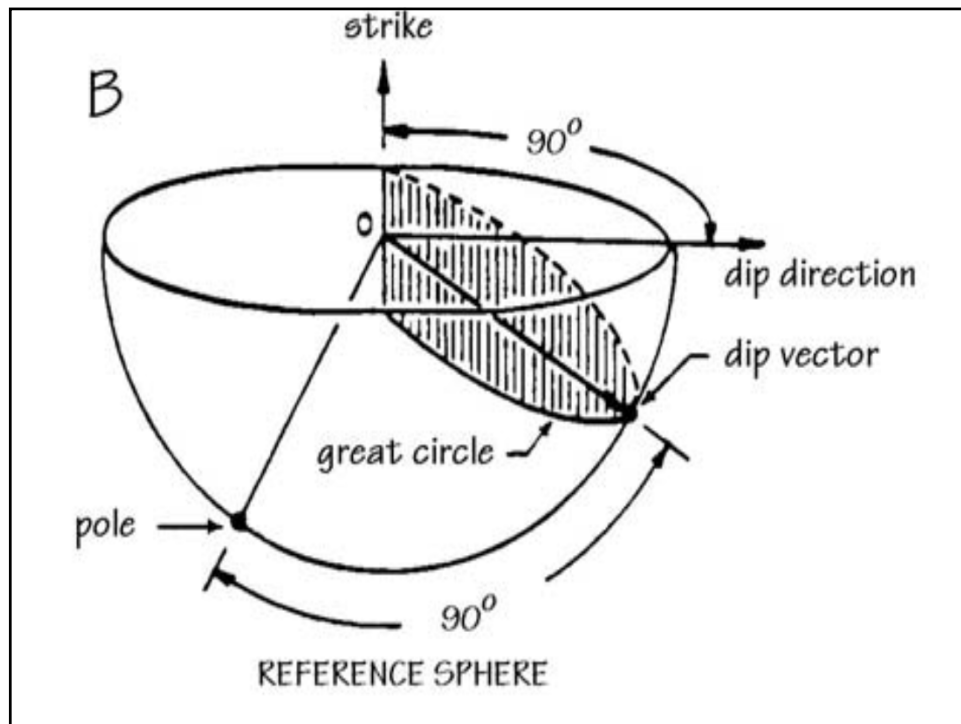
- the equal angle or Wulff net, and
- the equal area or Schmidt, Lambert net.

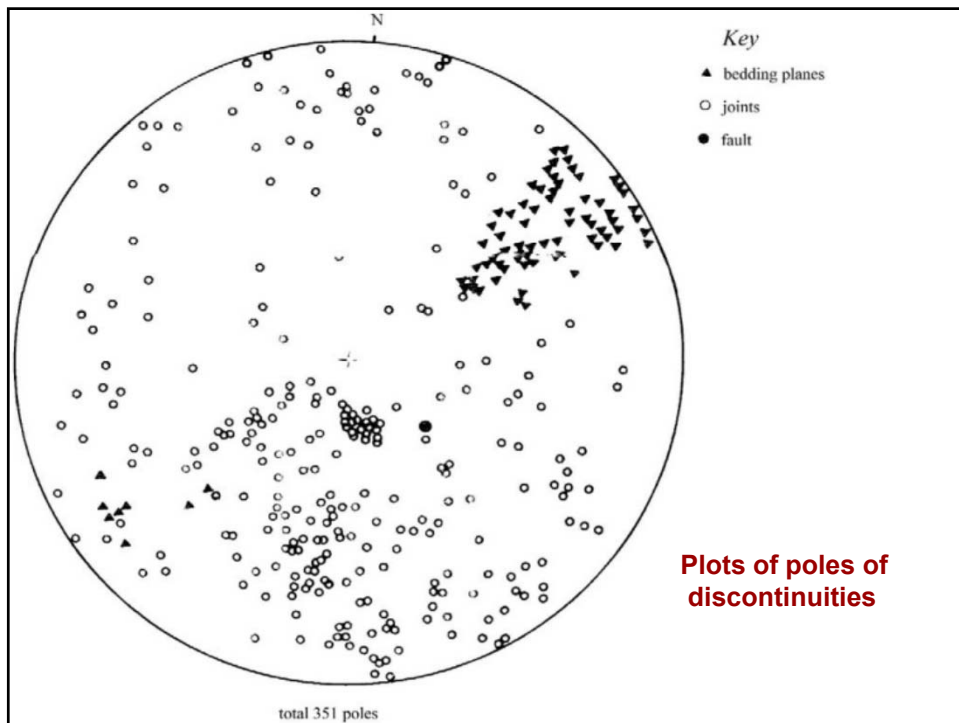
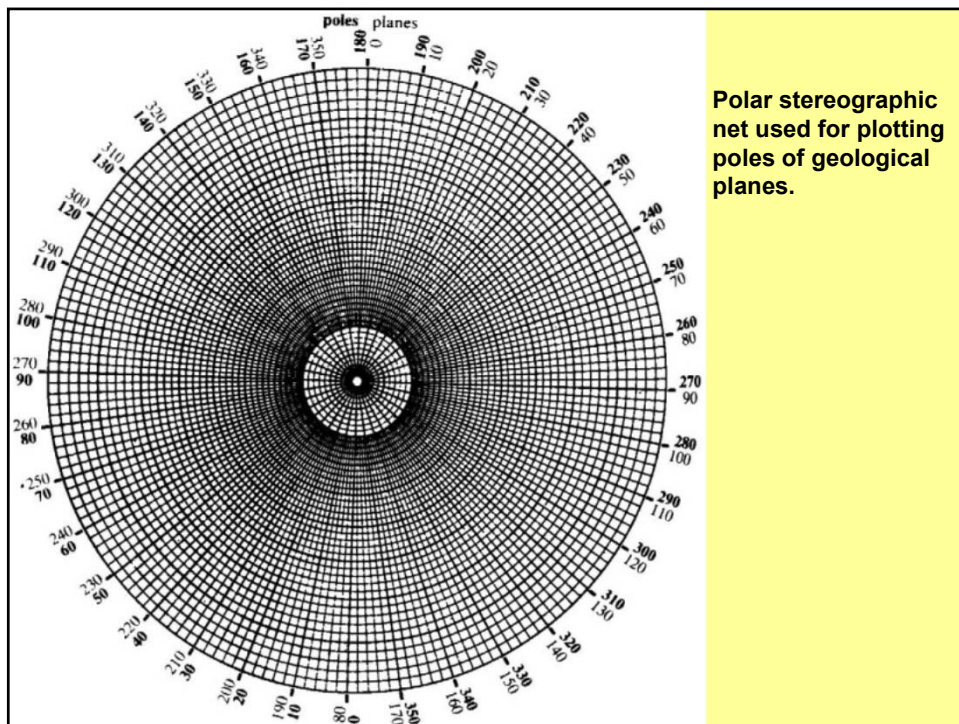
The equal angle projection is used in structural geology when angular relationships between geologic structures, such as bedding planes are being examined. The equal area projection is most often used when the distribution of planes within certain areas of the reference sphere is examined.

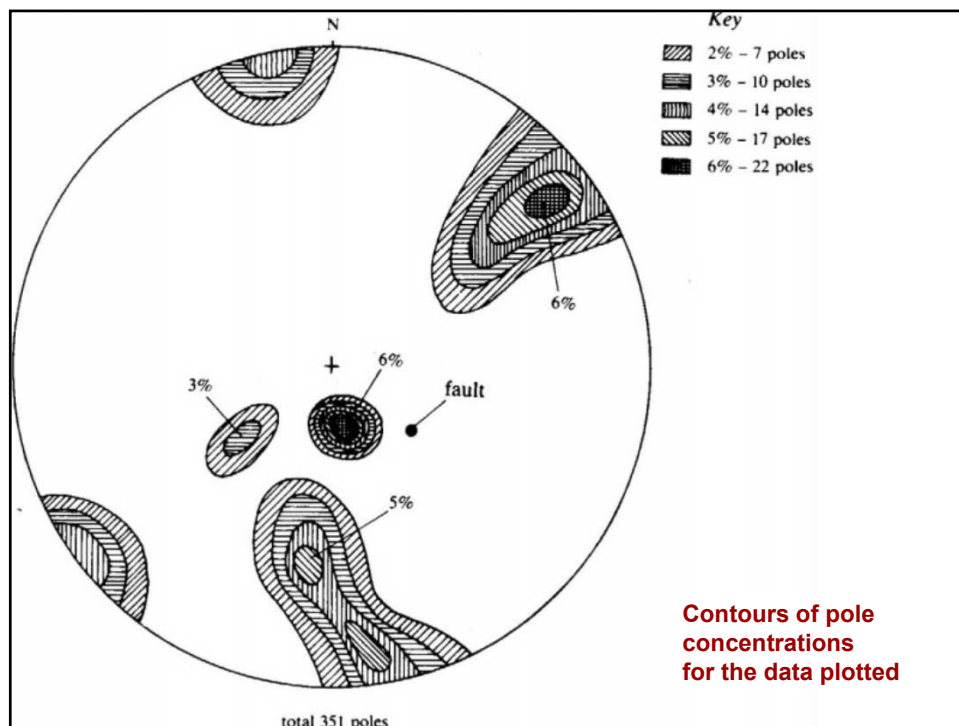
Plotting and analysis of discontinuity orientation data

In stereonet analyses, discontinuities are assumed to be planar. There are three possible representations of a plane in space on stereonet. They are **poles**, **dip vectors**, and **great circles**, as illustrated in Figure. However, it is slightly more convenient to use a suitably annotated **polar net**.









ROCK STRESS

Any attempt to design engineering structures in rock mass requires knowledge of the stress field.

The prior knowledge of the prevailing stress field and the stress which may develop due to excavation helps in the construction, alignment, stability analysis and the design of support system for the underground.